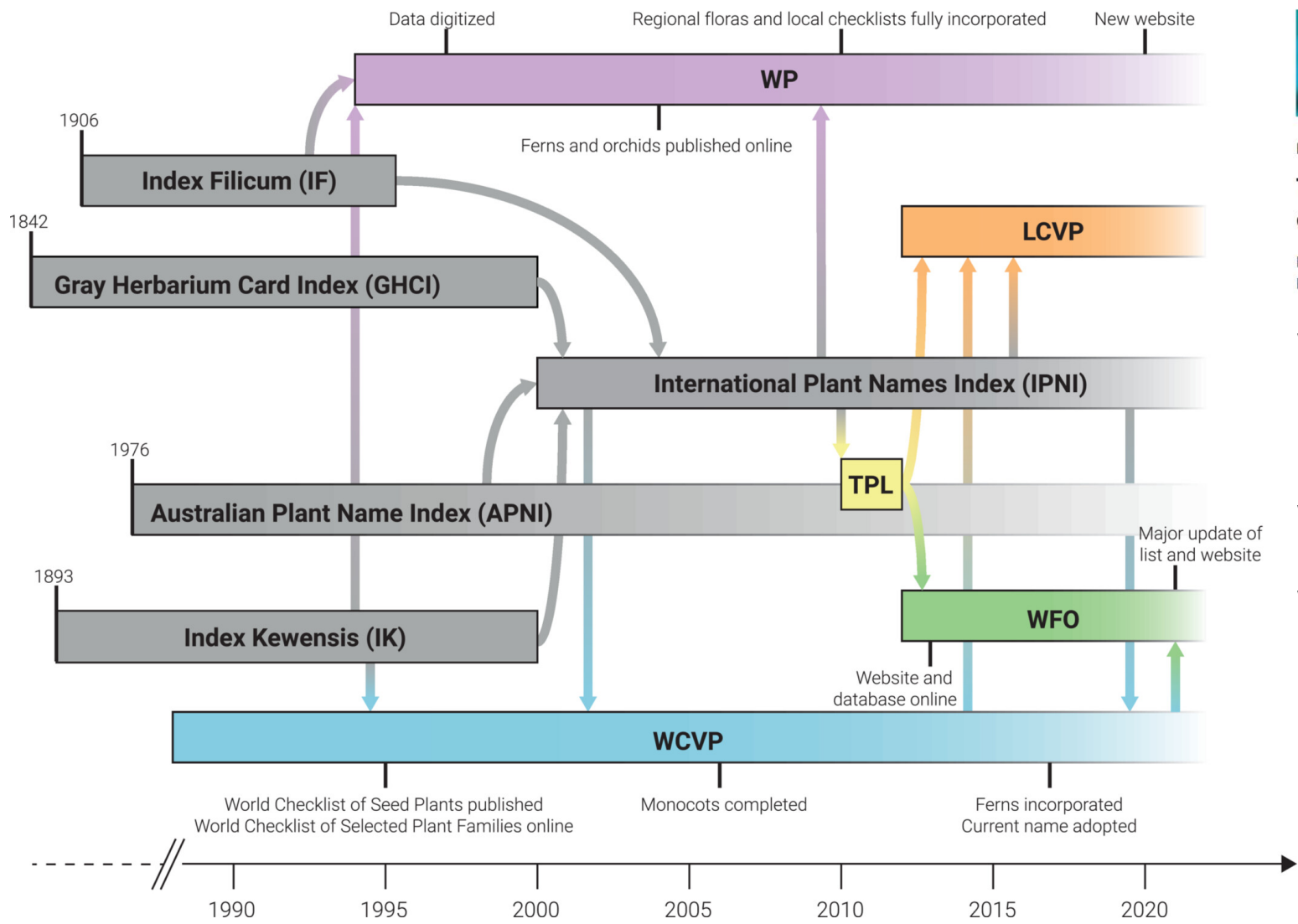


# **Bloque 2**

Herramientas en R para validación taxonómica

# CHECKLIST HISTORY





## New Phytologist

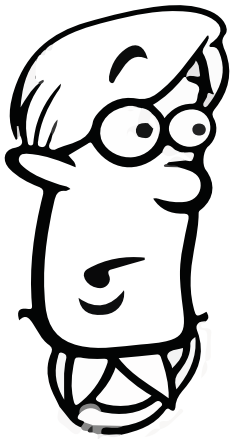
Methods | [Open Access](#) | [CC](#) [BY](#) [NC](#) [ND](#)

### The big four of plant taxonomy – a comparison of global checklists of vascular plant names

David Schellenberger Costa✉, Gerhard Boehnisch, Martin Freiberg, Rafaël Govaerts, Matthias Grenié, Michael Hassler, Jens Kattge, Alexandra N. Muellner-Riehl, Blanca M. Rojas Andrés ... [See all authors](#)

- WP – WorldPlants
- LCVP – Leipzig Catalogue of Vascular Plants
- WFO – World Flora Online
- WCVF – World Checklist of Vascular Plants





*Anagallis arabica* Duby

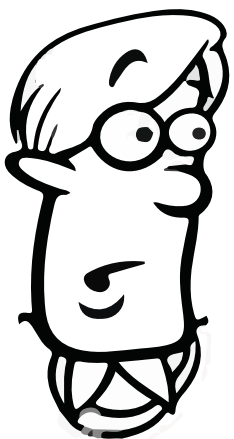


*Lysimachia arvensis* (L.) U.Manns & Anderb.



*Anagallis arvensis* L.





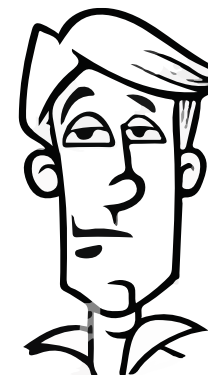
*Anagallis arabica* Duby



*Lysimachia arvensis* (L.) U.Manns & Anderb.



*Anagallis arvensis* L.





# TNRS

Taxonomic Name Resolution  
Service

An online tool for the standardization of plant  
taxonomic names.



# GBIF

Global Biodiversity  
Information Facility



Plant Diversity

Volume 45, Issue 1, January 2023, Pages 1-5



Short communication

## U.Taxonstand: An R package for standardizing scientific names of plants and animals

Jian Zhang <sup>a</sup>  , Hong Qian <sup>b</sup>  

### Package flora

### Description

Collect data from the Brazilian Flora checklist



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KeAi  
CHINESE ROOTS  
GLOBAL IMPACT

Plant Diversity

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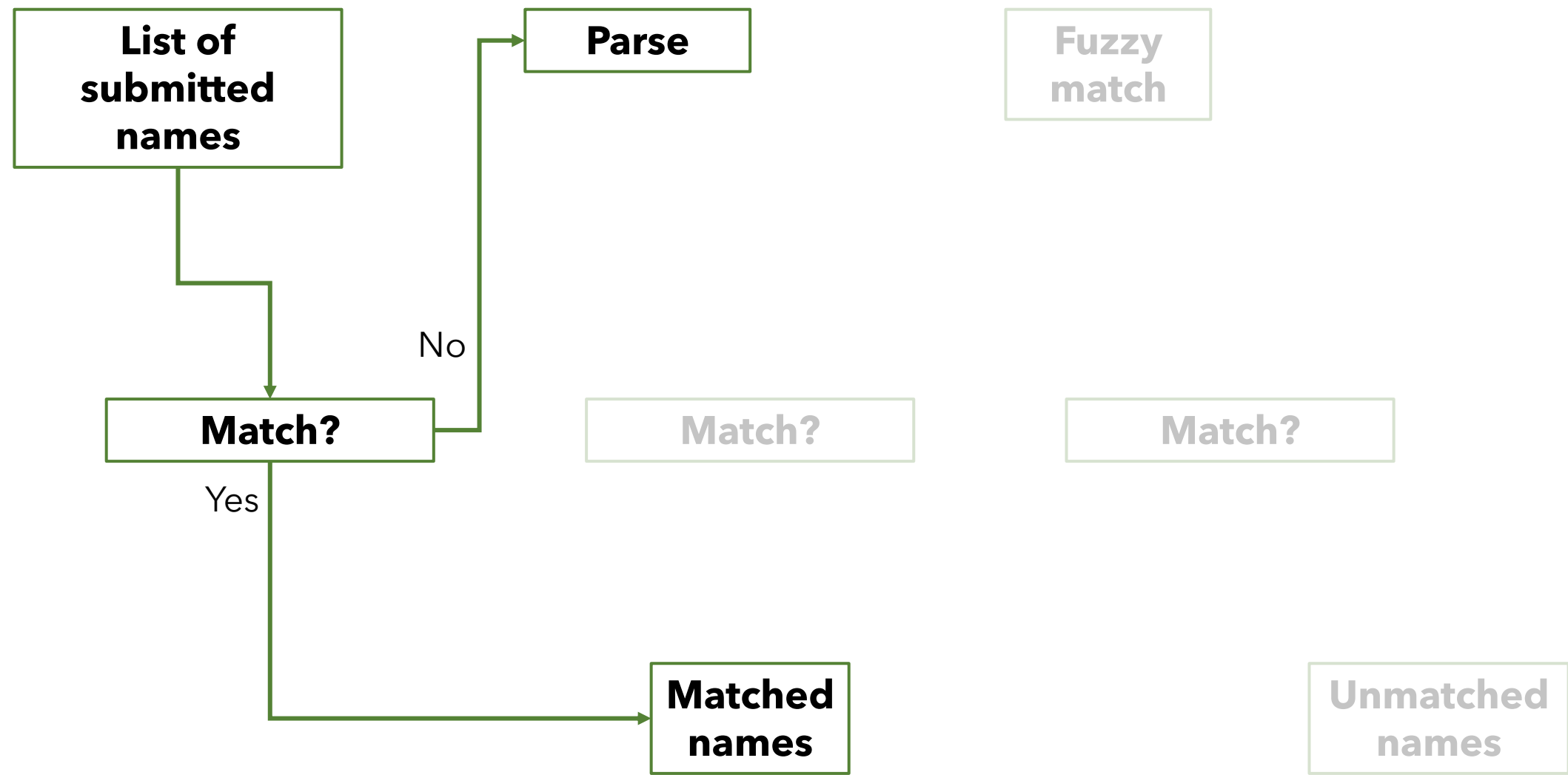
**LOCAL TOOL**



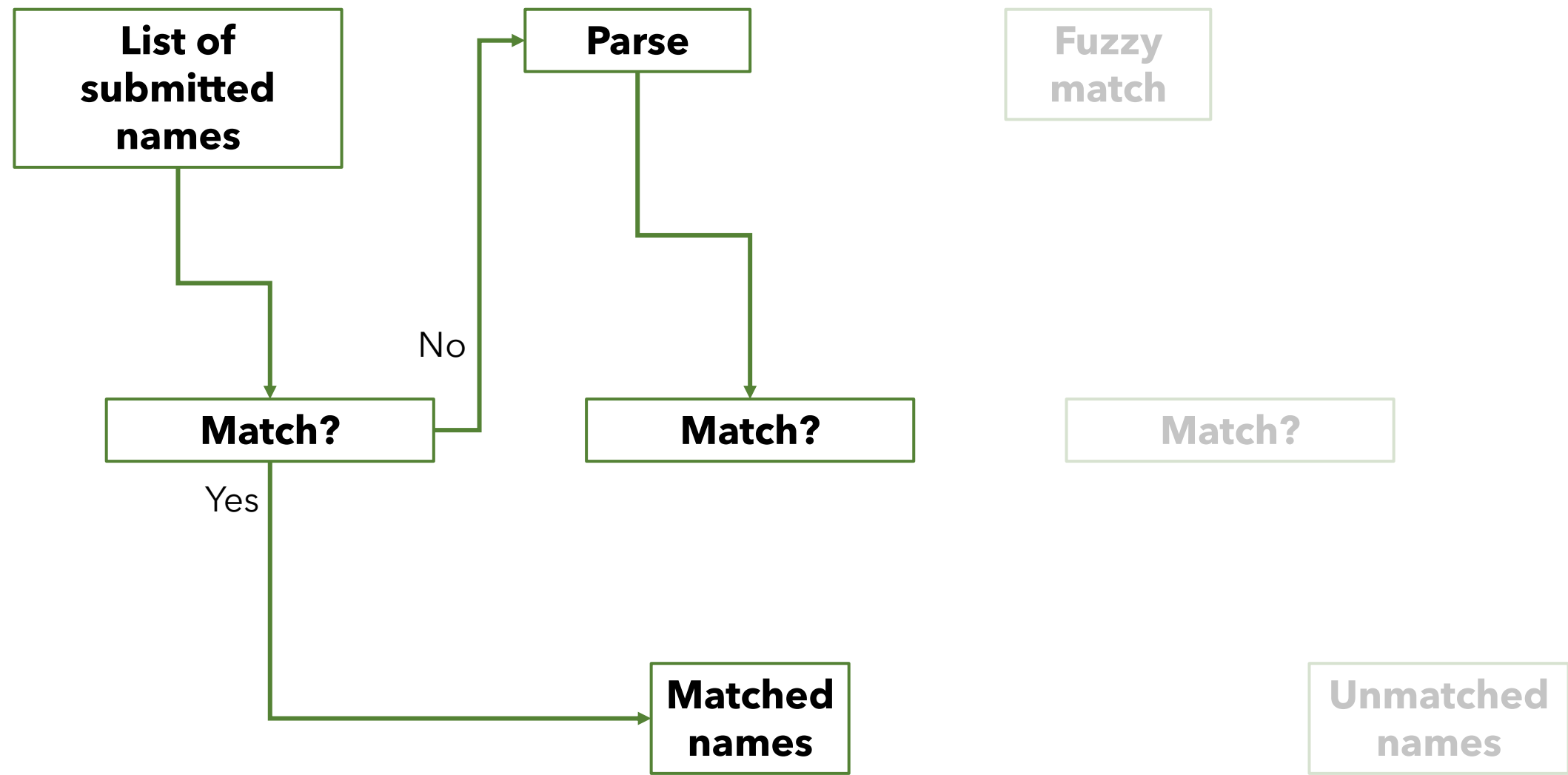
# MULTI-STEP DECISION PROCESS



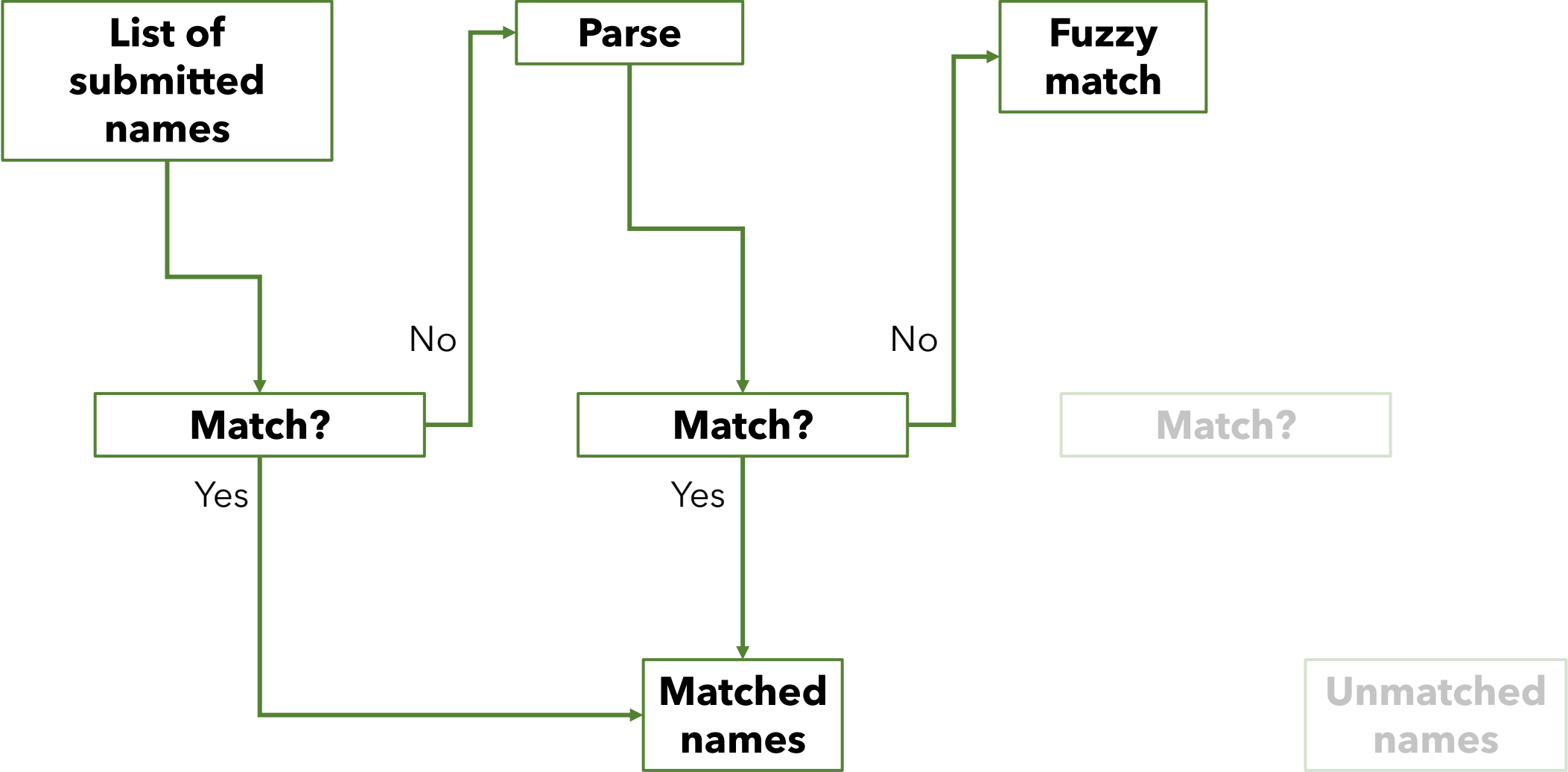
# MULTI-STEP DECISION PROCESS



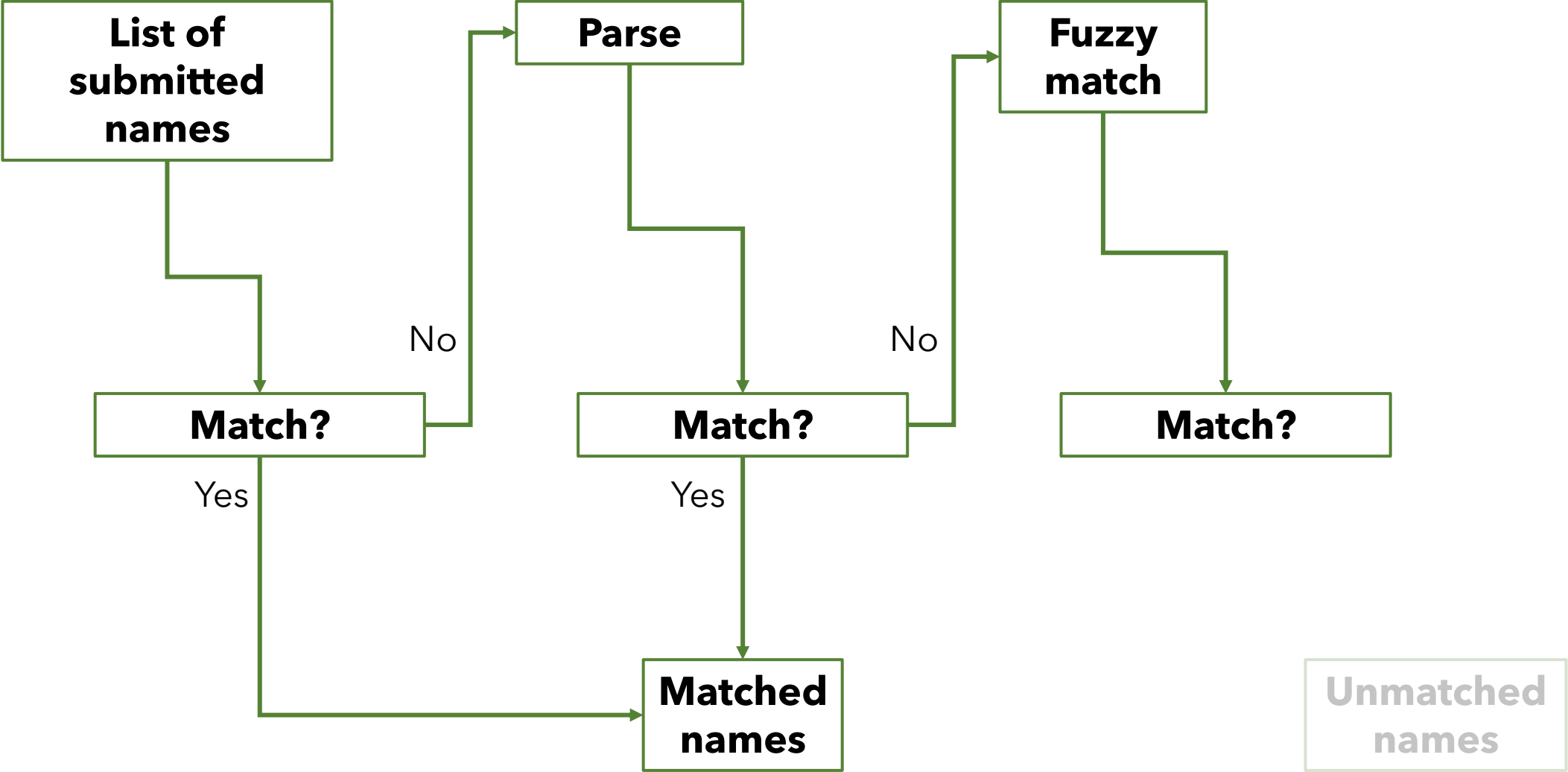
# MULTI-STEP DECISION PROCESS



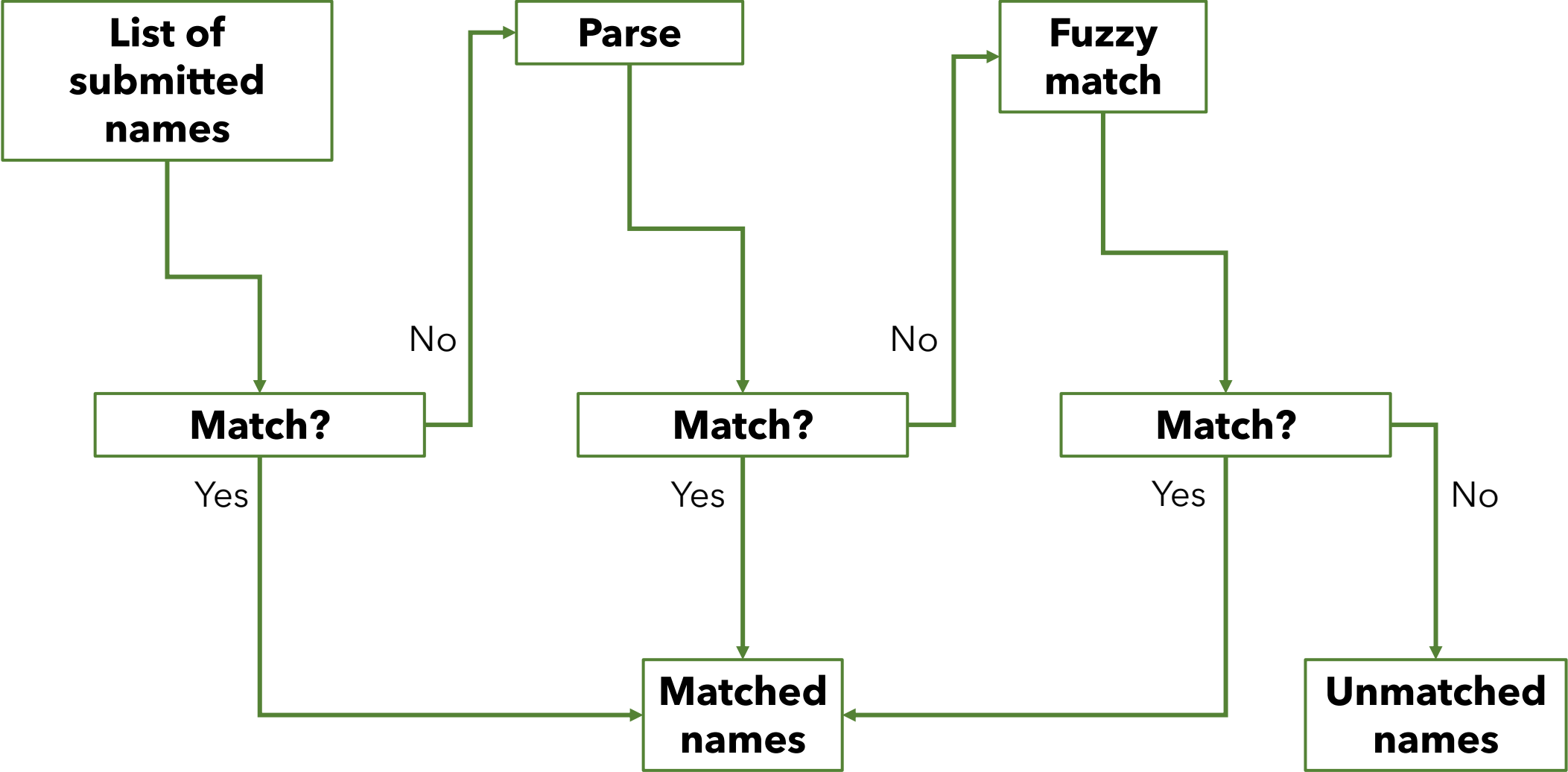
# MULTI-STEP DECISION PROCESS



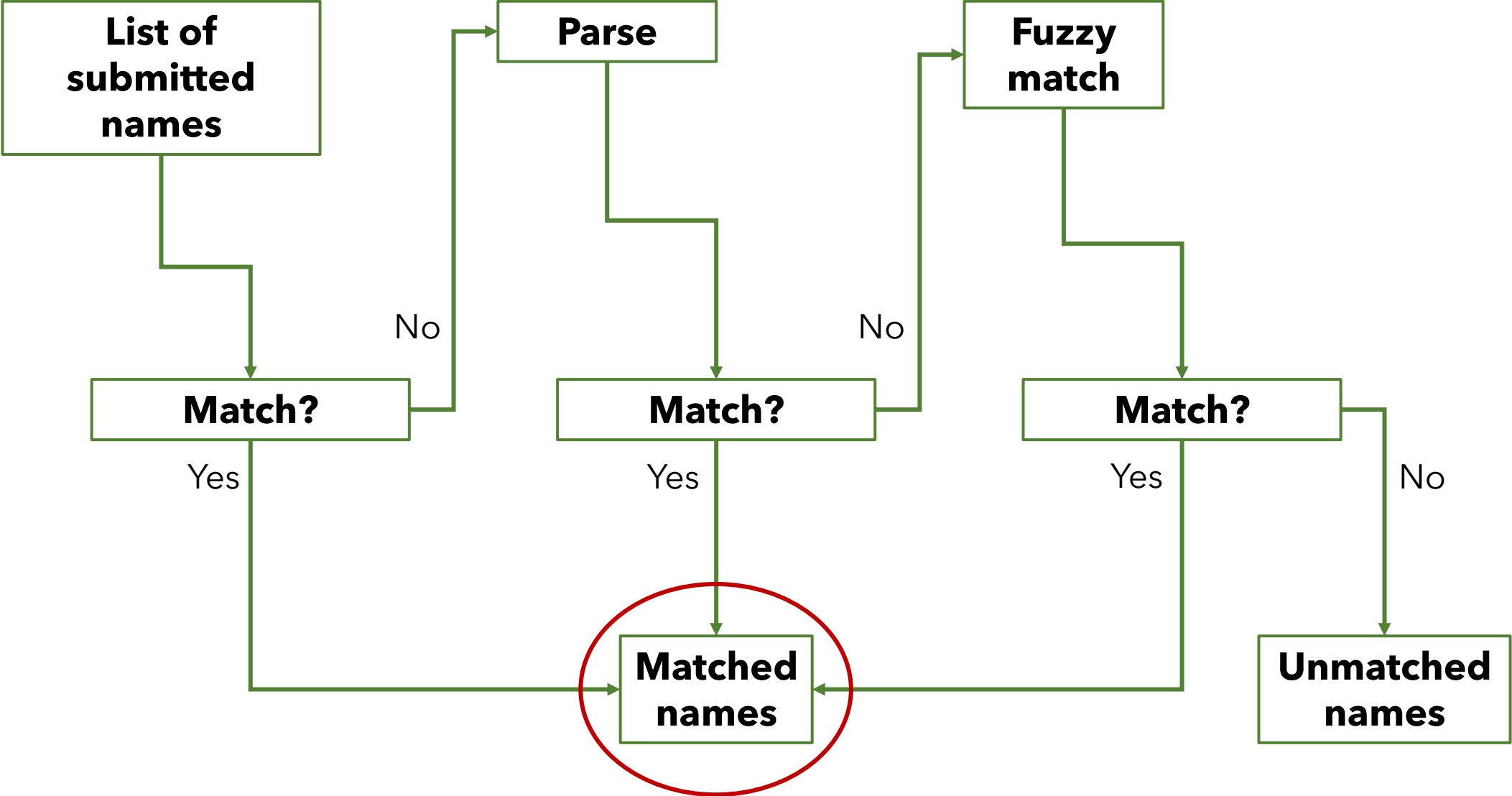
# MULTI-STEP DECISION PROCESS



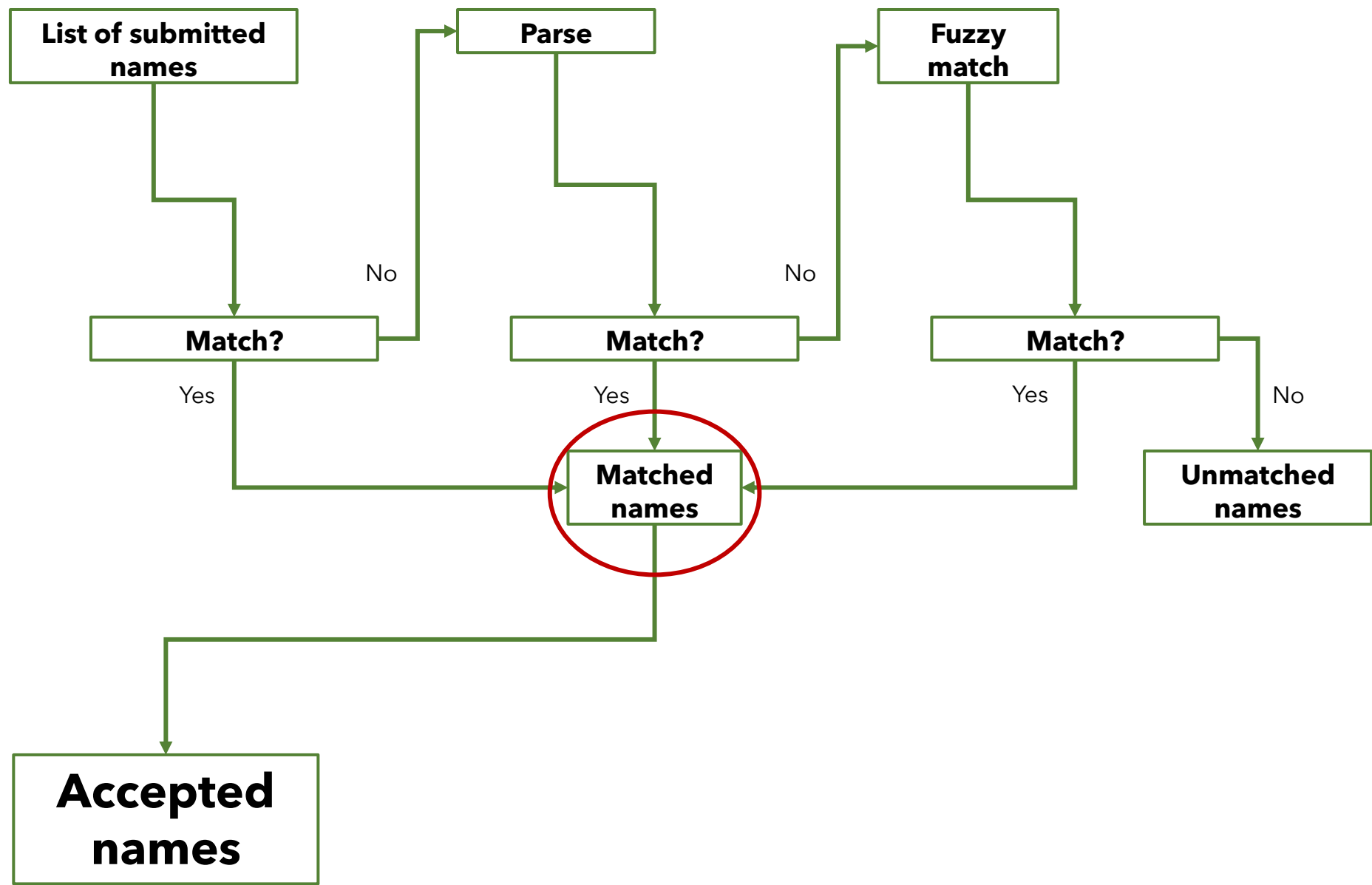
# MULTI-STEP DECISION PROCESS



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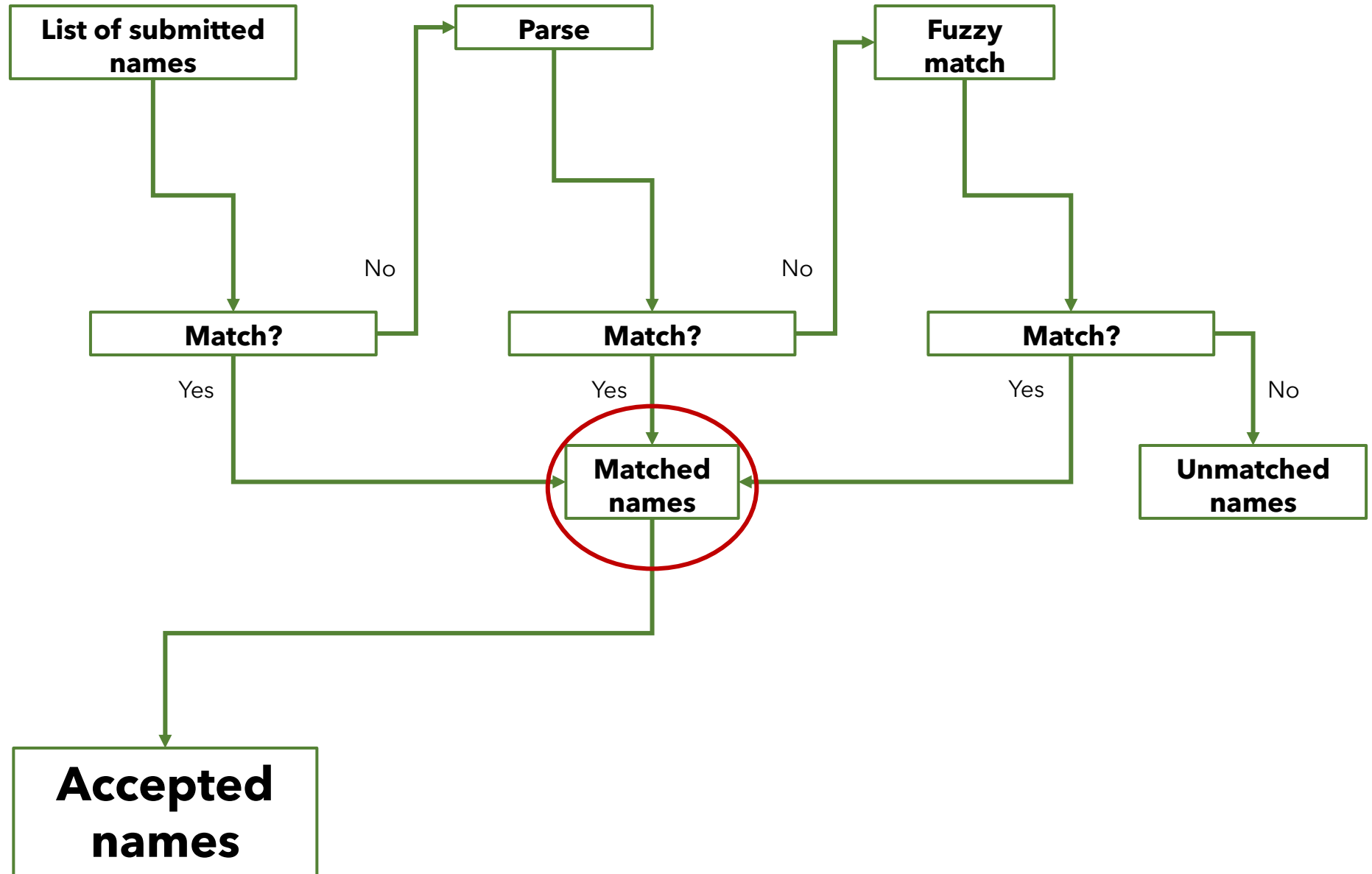


# MULTI-STEP DECISION PROCESS





# MULTI-STEP DECISION PROCESS





## **PROS**

- Online tool (Updated)
- Rely on WCVP or WFO
- Good fuzzy match

## **CONS**

- Rely on Server speed
- Checklist dependent



## **PROS**

- Online tool (Updated)
- Good fuzzy match

## **CONS**

- Rely on Server speed
- Uses GBIF Backbone Checklist

**Package flora**

**Description**

Collect data from the Brazilian Flora checklist

## **PROS**

- Online tool (Updated)
- Good for tropical species

## **CONS**

- Rely on Server speed
- Only on tropical species



Short communication

## U.Taxonstand: An R package for standardizing scientific names of plants and animals

Jian Zhang <sup>a</sup>  , Hong Qian <sup>b</sup>  

## PROS

- Easy to use
- User select the checklist

## CONS

- Rely on PC “speed”
- Updates manually

# HOW?



# GITHUB REPOSITORY

|                                                                                                                            |                                                                                                      |                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  <b>efernandezpascual</b> giacomo changes | dbb907c · yesterday                                                                                  |  <b>25 Commits</b> |
|  data                                     | giacomo changes                                                                                      | yesterday                                                                                             |
|  figures                                  | readme                                                                                               | yesterday                                                                                             |
|  results                                  | giacomo changes                                                                                      | yesterday                                                                                             |
|  src                                      | post meeting 20251125                                                                                | yesterday                                                                                             |
|  .gitignore                               | initial commit    | 2 weeks ago                                                                                           |
|  README.Rmd                              | readme                                                                                               | yesterday                                                                                             |
|  README.md                              | readme                                                                                               | yesterday                                                                                             |
|  nomenclatura.Rproj                     | initial commit  | 2 weeks ago                                                                                           |

# GITHUB REPOSITORY

|                                                                                                                            |                                                                                                      |                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  <b>efernandezpascual</b> giacomo changes | dbb907c · yesterday                                                                                  |  <b>25 Commits</b> |
|  data                                     | giacomo changes                                                                                      | yesterday                                                                                             |
|  figures                                  | readme                                                                                               | yesterday                                                                                             |
|  results                                  | giacomo changes                                                                                      | yesterday                                                                                             |
|  src                                      | post meeting 20251125                                                                                | yesterday                                                                                             |
|  .gitignore                               | initial commit    | 2 weeks ago                                                                                           |
|  README.Rmd                              | readme                                                                                               | yesterday                                                                                             |
|  README.md                              | readme                                                                                               | yesterday                                                                                             |
|  nomenclatura.Rproj                     | initial commit  | 2 weeks ago                                                                                           |



# GITHUB REPOSITORY

|                                                                                                      |                      |  |           |
|------------------------------------------------------------------------------------------------------|----------------------|--|-----------|
|  ..                 |                      |  |           |
|  1_Campo.csv        | Add files via upload |  | yesterday |
|  2_Catalogo.csv     | Add files via upload |  | yesterday |
|  3_Metaanalyses.csv | Add files via upload |  | yesterday |
|  table_test.RData   | Add files via upload |  | yesterday |

# GITHUB REPOSITORY

 **efernandezpascual** giacomo changes

dbb907c · yesterday 25 Commits

|                                                                                                        |                       |             |
|--------------------------------------------------------------------------------------------------------|-----------------------|-------------|
|  data                 | giacomo changes       | yesterday   |
|  figures              | readme                | yesterday   |
|  results              | giacomo changes       | yesterday   |
|  src                  | post meeting 20251125 | yesterday   |
|  .gitignore           | initial commit 🌱      | 2 weeks ago |
|  README.Rmd           | readme                | yesterday   |
|  README.md            | readme                | yesterday   |
|  nomenclatura.Rproj | initial commit 🌱      | 2 weeks ago |

 R\_script.R

post meeting 20251125

yesterday



# GITHUB REPOSITORY

|                                                                                                                            |                                                                                                      |                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  <b>efernandezpascual</b> giacomo changes | dbb907c · yesterday                                                                                  |  <b>25 Commits</b> |
|  data                                     | giacomo changes                                                                                      | yesterday                                                                                             |
|  figures                                  | readme                                                                                               | yesterday                                                                                             |
|  results                                  | giacomo changes                                                                                      | yesterday                                                                                             |
|  src                                      | post meeting 20251125                                                                                | yesterday                                                                                             |
|  .gitignore                               | initial commit    | 2 weeks ago                                                                                           |
|  README.Rmd                              | readme                                                                                               | yesterday                                                                                             |
|  README.md                              | readme                                                                                               | yesterday                                                                                             |
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# CLEAN FUNCTION

```
clean_name <- function(x){ x %>% str_squish() %>%  
  str_replace_all("\\s+", " ") %>% str_replace_all("(?i)\\b(sp|aff|nr|cf|spp|group|gr)\\.?.\\b", "")  
  %>% str_replace_all("[?]", "") %>% str_squish()  
}
```

## INPUT

Achillea sp

Carhtamus lanatus cf

## OUTPUT

Achillea

Carhtamus lanatus

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  str_replace_all("\\s+", " ") %>% str_replace_all("(?i)\\b(sp|aff|nr|cf|spp|group|gr)\\.?.\\b", "")  
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**INPUT**

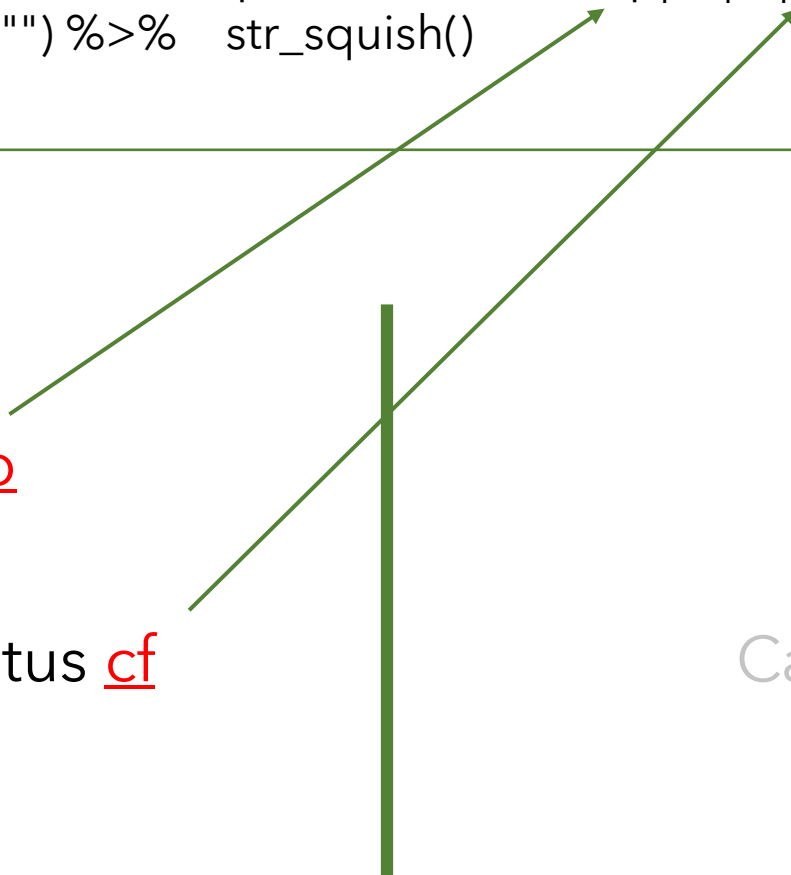
Achillea sp

Carhtamus lanatus cf

**OUTPUT**

Achillea

Carhtamus lanatus



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Carhtamus lanatus cf

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